# MITS 5300G – Assignment 1

**Instructions:**

- **Submit your answers in a single document (named "Lastname_firstname", in PDF format) through Canvas. It is your responsibility to verify that the assignment was properly submitted through Canvas by the deadline.**
- **Any references to "your Linux VM" refer to the Linux VM that is described in the "setup" section below.**
- **Please ensure you are familiar with the course policy on academic integrity. This assignment is to be completed and written up individually.**
- **Watch your spelling and grammar; you will lose marks for spelling and grammar mistakes.**
- **Keep all answers short, clear, and concise. Writing more sentences than necessary will not result in a better mark (and may result in deductions of marks)!**

**Setup:**

1. **Install VirtualBox or VMWare, and then create a new VM using the latest Fedora Security Labs ISO.**
2. **Once you have your Fedora VM created, you should ensure you create a snapshot of your VM for backup purposes.**

**Questions:**

1. (5.5 points) Set up your installed Fedora Linux VM to have two users, who are not members of the same group: *Sarah_<YourFullName>* and *Evelyn_<YourFullName>*, where <YourFullName> is replaced by your actual full name (e.g., this course's instructor would use "JulieThorpe"). Ensure that *Sarah_<YourFullName>*'s home directory has the permissions of "drwx-----x" and contains a file called "codes.txt". The "codes.txt" file has the permissions of "-rw----r--".

    a. (1 point) Submit a screenshot that shows *Evelyn_<YourFullName>'s* shell prompt, and the commands she must enter into her shell to ensure that:

        i. *Evelyn_<YourFullName>* cannot read the file, and

        ii. *Evelyn_<YourFullName>* cannot verify that the file exists.

    b. (4 points) Double-check that you have set up the scenario from part (a) on your Linux VM, such that *Evelyn_<YourFullName>* does not have access to *Sarah_<YourFullName>*'s codes.txt. Attach the CD/DVD drive of your VM to a Linux Live iso file (e.g., the Fedora

Security Lab iso you also used for VM setup, or another distribution you have). Reboot the machine into the Live OS (you can verify this by checking the usernames available in /etc/passwd --- if using the Fedora Security Lab iso, you should see 'liveuser'). If it were Evelyn (the person who is associated with *Evelyn_<YourFullName>*) rebooting the system into this Live OS, what access could she have to *Sarah_<YourFullName>*'s codes.txt? Keep in mind that Evelyn is very determined to access *Sarah_<YourFullName>*'s files and will use all features offered by the Live OS. Explain in 2-3 sentences, and back up your answer with 1-2 screen shots of what access *Evelyn_<YourFullName>* could have. Make sure you include at least one screenshot that shows the file permissions of a file listed as belonging to *Sarah_<YourFullName>* (note that in your Live CD, you may only see a UID rather than a username---remember that you can show who that UID is by showing a screenshot of the account in your /etc/passwd file).

c. (0.5 points) Consider the protection provided by encrypting confidential files. Imagine that Sarah had an account on a multi-user Linux system, and encrypted all of her files using her password, but did not set the permissions to her confidential files and/or directories that house them. Briefly describe one attack against Sarah's files in this scenario that should concern her.

2. (2.5 points) On a Windows system, install *Process Monitor*, available for download here: http://technet.microsoft.com/en-us/sysinternals/bb896645 . Explore the filtering options to see what sort of registry activity you can focus collection/analysis efforts upon.

- (0.5 points) Provide a reference for, and briefly describe an attack scenario that involves the use or modification of the Windows registry.

- (2 points)
    i. Describe how you could use and set up filters for *Process Monitor* that would have worked to detect the attack scenario you describe in (a), if it were running at an appropriate time to detect it. Make sure you describe all of the filter options.

    ii. Add a ProcessMonitor filter to exclude a "ProcessName" that is "YourFullName" (e.g., this course's professor would use "JulieThorpe") and then take at least one screenshot that shows this filter and the filters you would set up according to (i).